NEWTON SCHOOL

ALGEBRA 1 | ANSWER KEY

Rational Expressions & Equations — Test Redo Mix

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Standard: CCSS.MATH.CONTENT.HSA-APR.D.6 & HSA-REI.A.2

Part A — SIMPLIFY

A-1 · Linear Simplify

Problem 1

Simplify and state restriction: $\dfrac{a-b}{a^2-b^2}$

Difference of Squares | Cancel | Restriction

Step 1. Factor denominator: $a^2-b^2=(a-b)(a+b)$.

Step 2. $\dfrac{\cancel{(a-b)}}{\cancel{(a-b)}\,(a+b)}=\dfrac{1}{a+b}$.

Step 3 — Restriction: original denominator zero when $a=b$ or $a=-b$. After cancel we still need $a\neq b$ (lost factor) AND $a\neq -b$.

$\boxed{\dfrac{1}{a+b},\quad a\neq b,\quad a\neq -b}$

Problem 2

Simplify: $\dfrac{y}{x-y}+\dfrac{x}{y-x}$

Opposites | Common Denom

Step 1. Rewrite $(y-x)=-(x-y)$: $\dfrac{x}{y-x}=\dfrac{x}{-(x-y)}=-\dfrac{x}{x-y}$.

Step 2. $\dfrac{y}{x-y}-\dfrac{x}{x-y}=\dfrac{y-x}{x-y}$.

Step 3. $\dfrac{y-x}{x-y}=\dfrac{-(x-y)}{x-y}=-1$.

$\boxed{-1, x\neq y}$

Problem 3

Simplify: $\dfrac{3}{x-3} - \dfrac{x}{3-x}$

Opposites | Common Denom

Step 1. $(3-x) = -(x-3)$, so $\dfrac{x}{3-x} = -\dfrac{x}{x-3}$.

Step 2. $\dfrac{3}{x-3} - \left(-\dfrac{x}{x-3}\right) = \dfrac{3}{x-3} + \dfrac{x}{x-3}$.

Step 3. $= \dfrac{3+x}{x-3} = \dfrac{x+3}{x-3}$.

$$\boxed{\frac{x+3}{x-3}, \quad x \neq 3}$$

A-2 · Quadratic Simplify

Problem 4

Simplify: $\dfrac{9m^2-4}{3m^2+4m-4}$

Difference of Squares | Factor Trinomial (AC method)

Step 1. Numerator: $9m^2 - 4 = (3m-2)(3m+2)$.

Step 2. Denominator $3m^2+4m-4$: AC $= 3 \cdot (-4) = -12$. Split $4m = 6m - 2m$:
$3m^2 + 6m - 2m - 4 = 3m(m+2) - 2(m+2) = (3m-2)(m+2)$.

Step 3. Cancel $(3m-2)$: $\dfrac{\cancel{(3m-2)}\,(3m+2)}{\cancel{(3m-2)}\,(m+2)}$.

$$\boxed{\frac{3m+2}{m+2}, \quad m \neq \tfrac{2}{3}, \quad m \neq -2}$$

Problem 5

Simplify: $\dfrac{x^2-4}{x^2-x-6} \div \dfrac{x^2+4x+4}{x^2-9}$

Division → Reciprocal | Factor All

Step 1. Multiply by reciprocal: $\dfrac{x^2-4}{x^2-x-6} \cdot \dfrac{x^2-9}{x^2+4x+4}$.

Step 2. Factor every polynomial: $\dfrac{(x-2)(x+2)}{(x-3)(x+2)} \cdot \dfrac{(x-3)(x+3)}{(x+2)^2}$.

Step 3. Cancel $(x+2)$ once and $(x-3)$: result is $\dfrac{(x-2)(x+3)}{(x+2)^2}$.

$$\boxed{\frac{(x-2)(x+3)}{(x+2)^2}}$$

Problem 6 (bonus-style)

Simplify: $\left(y - \dfrac{2}{y-1}\right)\left(y + \dfrac{1}{y-2}\right)$

Combine Fractions Factor Trinomials Cancel

Step 1. First factor (common denom $y-1$): $\dfrac{y(y-1)-2}{y-1} = \dfrac{y^2-y-2}{y-1} = \dfrac{(y-2)(y+1)}{y-1}$.

Step 2. Second factor (common denom $y-2$): $\dfrac{y(y-2)+1}{y-2} = \dfrac{y^2-2y+1}{y-2} = \dfrac{(y-1)^2}{y-2}$.

Step 3. Multiply and cancel: $\dfrac{(y-2)(y+1)}{\cancel{(y-1)}} \cdot \dfrac{\cancel{(y-1)}\,(y-1)}{\cancel{(y-2)}} = (y+1)(y-1)$.

$\boxed{y^2-1}$

Part B — SOLVE

B-1 · Linear Solve

Problem 7

Solve: $\dfrac{3}{x+2} = \dfrac{1}{x-2}$

Cross-Multiply · Check

Step 1. Cross-multiply: $3(x-2) = 1(x+2)$.

Step 2. $3x - 6 = x + 2 \;\Rightarrow\; 2x = 8 \;\Rightarrow\; x = 4$.

Step 3 — Check. $x + 2 = 6,\; x - 2 = 2$ — both non-zero. Valid.

$\boxed{x = 4}$

Problem 8

Solve: $\dfrac{x-1}{x+3} = 2$

Clear Denom · Linear

Step 1. Multiply both sides by $x+3$: $x - 1 = 2(x+3)$.

Step 2. $x - 1 = 2x + 6 \;\Rightarrow\; -7 = x$.

Step 3 — Check. $x + 3 = -4 \neq 0$. Valid.

$\boxed{x = -7}$

Problem 9

Solve: $\dfrac{2}{x-1} + \dfrac{3}{x-1} = \dfrac{5}{2}$

Combine LHS · Cross-Multiply

Step 1. Combine: $\dfrac{2+3}{x-1} = \dfrac{5}{x-1}$.

Step 2. $\dfrac{5}{x-1} = \dfrac{5}{2} \;\Rightarrow\; x - 1 = 2 \;\Rightarrow\; x = 3$.

Step 3 — Check. $x - 1 = 2 \neq 0$. Valid.

$\boxed{x = 3}$

B-2 · Quadratic Solve

Problem 10

Solve: $\dfrac{x}{x-1} - \dfrac{2}{x+1} = \dfrac{4}{x^2-1}$

Clear Denom | Quadratic | Check Extraneous

Step 1. $x^2-1=(x-1)(x+1)$. Multiply everything by $(x-1)(x+1)$: $x(x+1)-2(x-1)=4$.

Step 2. $x^2+x-2x+2=4 \;\Rightarrow\; x^2-x-2=0$.

Step 3. $(x-2)(x+1)=0 \;\Rightarrow\; x=2$ or $x=-1$.

Step 4 — Check. $x=-1$ makes $x+1=0$. ⚠ x = -1 EXTRANEOUS

$\boxed{x=2}$

Problem 11

Solve: $\dfrac{1}{x} + \dfrac{1}{x+3} = \dfrac{1}{2}$

LCD | Quadratic

Step 1. LCD = $2x(x+3)$: $2(x+3)+2x=x(x+3)$.

Step 2. $2x+6+2x=x^2+3x \;\Rightarrow\; 4x+6=x^2+3x$.

Step 3. $x^2-x-6=0 \;\Rightarrow\; (x-3)(x+2)=0$.

Step 4 — Check. Neither $x=3$ nor $x=-2$ makes a denominator zero. Both valid.

$\boxed{x=3 \text{ or } x=-2}$

Problem 12

Solve: $\dfrac{2x}{x-3} = \dfrac{6}{x-3} + \dfrac{x}{2}$

Clear Denom | Quadratic | Check Extraneous

Step 1. Multiply by $2(x-3)$: $4x=12+x(x-3)$.

Step 2. $4x=12+x^2-3x \;\Rightarrow\; 0=x^2-7x+12$.

Step 3. $(x-3)(x-4)=0 \;\Rightarrow\; x=3$ or $x=4$.

Step 4 — Check. $x=3$ makes $x-3=0$. ⚠ x = 3 EXTRANEOUS

$\boxed{x=4}$